Jai Kulkarni

FPGA Design Engineer

RESUME_1393

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Total Experience: 10 years

PROFESSIONAL SUMMARY

FPGA Design Engineer with 10 years of experience in fpga, using VHDL, Verilog, Xilinx Vivado, Intel Quartus. Experienced in rtl simulation, fixed-point arithmetic and measurable delivery outcomes.

TECHNICAL SKILLS

Skills: Timing Closure, VHDL, Tcl, Verilog, FPGA, DSP, Xilinx Vivado, UART, Python, AXI4, Intel Quartus, SPI

Methods: RTL simulation, Fixed-point arithmetic, Board bring-up

PROFESSIONAL EXPERIENCE

Lead FPGA Design Engineer | TerraNex Solutions | Jan 2023 - Present

- Improved maximum operating frequency by 12% after timing-path optimization
- Reduced FPGA resource utilization by 31% through arithmetic sharing
- Lowered end-to-end data-path latency by 34% using pipelined logic
- Coordinated engineering design reviews, acceptance criteria and test evidence with cross-functional stakeholders.

FPGA Design Engineer | NovaPeak Engineering | Jan 2019 - Dec 2022

- Lowered end-to-end data-path latency by 32% using pipelined logic
- Verified 665 protocol corner cases with repeatable simulation suites

Associate FPGA Design Engineer | Evergrid Technologies | Jul 2016 - Dec 2018

- Improved maximum operating frequency by 29% after timing-path optimization
- Reduced FPGA resource utilization by 24% through arithmetic sharing

PROJECTS

Real-Time Signal Filter | AXI4, Intel Quartus, SPI, Python

- Implemented fixed-point FPGA pipelines with testbench golden models.
- Documented acceptance criteria and evaluated the design using fixed-point arithmetic and board bring-up.

Camera Data Bridge | Intel Quartus, Xilinx Vivado, AXI4, SPI

- Designed AXI streaming logic, frame buffers and clock-domain boundaries.
- Documented acceptance criteria and evaluated the design using fixed-point arithmetic and board bring-up.

EDUCATION

B.Tech in Electronics and Communication Engineering, Narmada School of Technology, 2016
Relevant Methods: RTL simulation, Fixed-point arithmetic, Board bring-up

CERTIFICATIONS

FPGA Architecture Workshop (2023)